

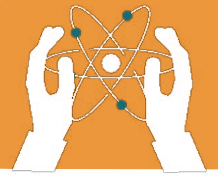
UNIT 4

AIR AND ITS COMPOSITION

Learning objectives:

- Identify the composition of air and the roles of its main gases (oxygen, nitrogen, carbon dioxide).
- Explain the importance of oxygen for respiration and combustion.
- Understand air pollution, its causes, and ways to reduce its impact.

AIR AND ITS COMPOSITION



4.1 Introduction to Air

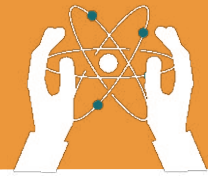
Air is an invisible mixture of gases essential for life. It supports breathing, fuels burning, and drives weather and the water cycle. The atmosphere, Earth's protective layer of air, shields us from harmful sun rays and maintains a stable temperature.



4.2 Gases in the Atmosphere

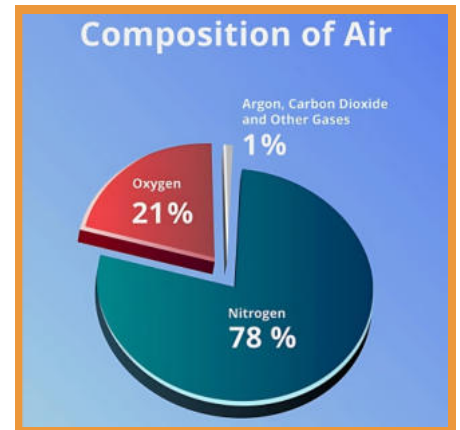
The air we breathe contains a mixture of gases:

- **Nitrogen:** Makes up about 78% of the air and helps plants grow.
- **Oxygen:** Makes up about 21% of the air and is vital for breathing and burning.



AIR AND ITS COMPOSITION

- **Carbon Dioxide:** A small percentage but essential for plants during photosynthesis.
- **Other Gases:** Argon, helium, and trace gases are present in very small amounts.

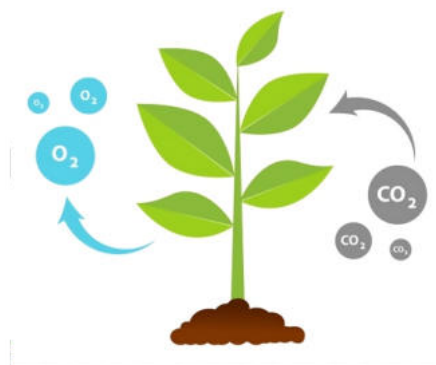


4.3 Importance of Oxygen

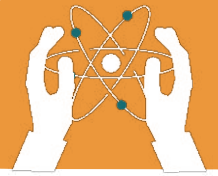
Oxygen is crucial for life as it is required for breathing. It combines with food in our bodies to release energy. This process is called **respiration**. Oxygen also supports combustion, enabling fuels to burn, which is essential for cooking, heating, and generating electricity.

4.4 Importance of Carbon Dioxide

Carbon dioxide is essential for plants as they use it in photosynthesis to make food. Although present in small amounts, it helps regulate Earth's temperature. Too much carbon dioxide, however, can contribute to global warming.

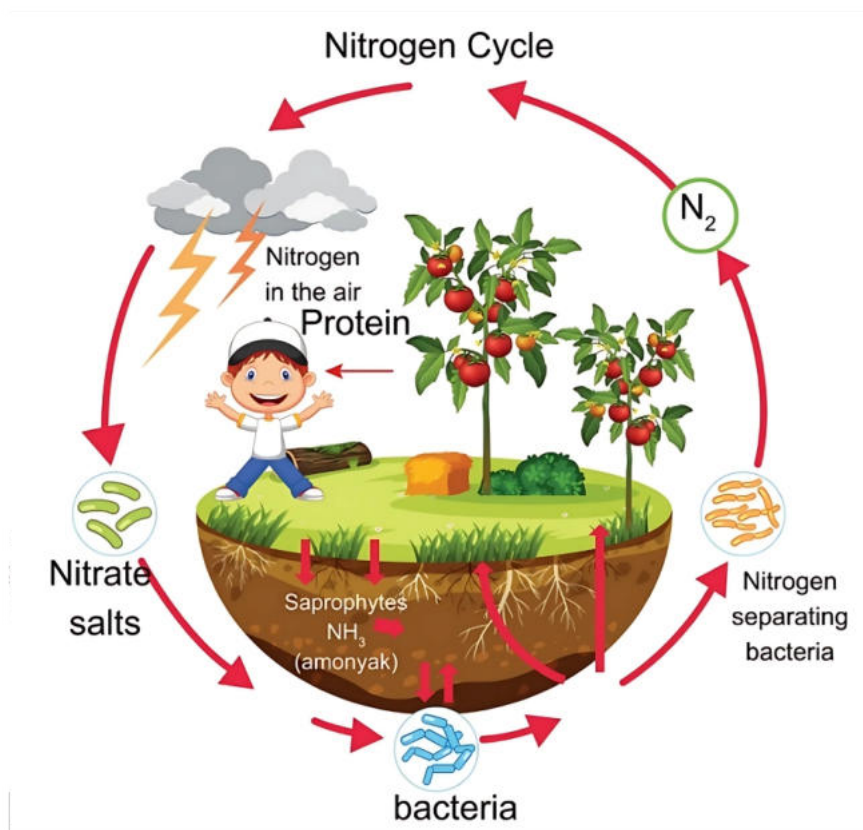


AIR AND ITS COMPOSITION



4.5 Importance of Nitrogen

Nitrogen is the most abundant gas in the air. Plants use nitrogen to grow, but they cannot absorb it directly from the air. Microorganisms in the soil convert nitrogen into forms that plants can use.



Do You Know?

Global warming is when the Earth's temperature gets hotter over time because of gases like carbon dioxide trapping heat in the atmosphere. It's caused by things like burning fuels and cutting down trees.

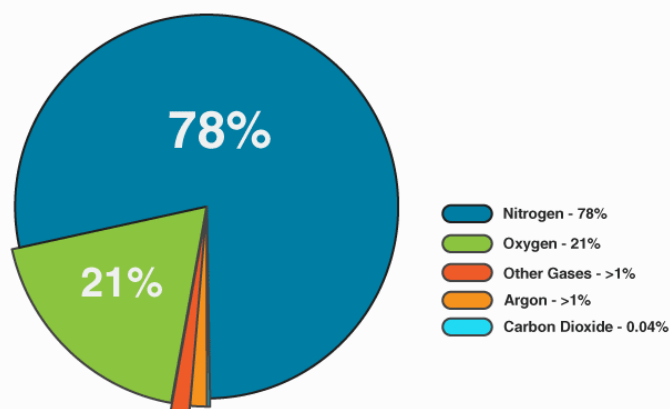


AIR AND ITS COMPOSITION

4.6 Other Gases in Air

Other gases, like argon, helium, neon, krypton, and xenon exist in small amounts in the air. While they do not play a major role in life processes, they are used in industries. For example, helium is used in balloons, and argon is used in welding.

COMPOSITION OF INHALED AIR



4.7 Properties of Air

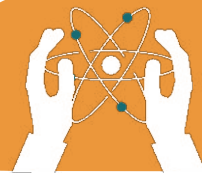
Air has the following properties:

- It occupies space.
- It has weight.
- It exerts pressure. These properties allow activities



like flying airplanes, sailing boats, and inflating tires.

AIR AND ITS COMPOSITION



4.8 Uses of Air

Air has many uses:

- Oxygen is used for breathing.
- Nitrogen is used in fertilizers.
- Carbon dioxide is used in fire extinguishers.
- Air is used in industries to generate energy.

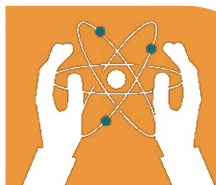
4.9 Air Pollution

Air pollution occurs when harmful gases, dust, and smoke mix with the air. It can cause health problems like asthma and harm the environment. Factories, vehicles, and burning waste contribute to pollution. Planting trees and using cleaner fuels can help reduce pollution.

• Causes of Air pollution

- Volcanic Eruptions release sulfur dioxide, ash, and particulate matter into the air.
- Dust Storms contribute to particulate matter, affecting air quality.
- Factories and power plants burn fossil fuels such as coal, oil, and natural gas, releasing pollutants like sulfur dioxide, nitrogen oxides, and particulate matter.
- Use of cleaning agents, paints, and sprays emits volatile organic compounds (VOCs), which are pollutants.





AIR AND ITS COMPOSITION

EXERCISES

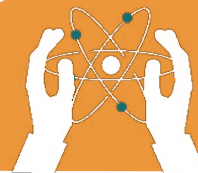
Fill in the blanks.

1. The most abundant gas in the atmosphere is _____ .
2. Plants use _____ for photosynthesis.
3. _____ is necessary for breathing.
4. Air _____ space and has weight.
5. _____ is a gas used in fire extinguishers.
6. Too much _____ in the air causes global warming.
7. _____ gas is used to inflate balloons.
8. Air pollution can be reduced by planting _____ .

Tick(✓) for True and cross (x) for False statements

1. Air is made up of only one gas.
2. Nitrogen makes up 78% of the air.
3. Plants absorb oxygen for photosynthesis.
4. Carbon dioxide is harmful to plants.
5. Oxygen is required for combustion.
6. Air occupies space and exerts pressure.
7. Argon is the most abundant gas in the air.
8. Planting trees helps reduce air pollution.

AIR AND ITS COMPOSITION



Choose the correct answer

1.The most abundant gas in the atmosphere is:

- a) Oxygen
- b) Carbon dioxide
- c) Nitrogen
- d) Argon

2. Plants use for photosynthesis:

- a) Oxygen
- b) Nitrogen
- c) Carbon dioxide
- d) Hydrogen

3.Air exerts :

- a) Pressure
- b) Weight
- c) Both a and b
- d) None of these

4. is used in fire extinguishers:

- a) Nitrogen
- b) Carbon dioxide
- c) Oxygen
- d) Helium

5.Which gas supports combustion?

- a) Helium
- b) Oxygen
- c) Nitrogen
- d) Argon

6.Air pollution is caused by:

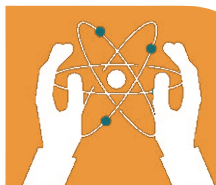
- a) Burning fuels
- b) Planting trees
- c) Cleaning the environment
- d) Using solar power

7.Helium is used for:

- a) Fertilizers
- b) Balloons
- c) Welding
- d) Cooling

8.Too much carbon dioxide leads to:

- a) Global warming
- b) Plant growth
- c) Clean air
- d) Acid rain



AIR AND ITS COMPOSITION

Answer the Following Questions:

Short-Answer Questions

1. What are the main gases in the air?
2. What is the role of carbon dioxide in plants?
3. How does nitrogen help plants?
4. What are the uses of air?

Long-Answer Questions

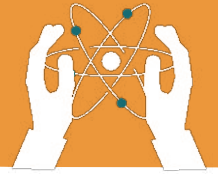
1. What are the effects of air pollution?

Reasoning Questions Answer

1. Why is oxygen important?
2. Why is air important for life?

Short Note

1. Write a short note on composition of air.



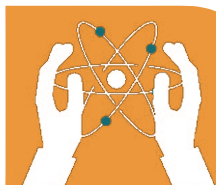
ACTIVITY

- Fill a balloon with air and observe its weight and volume. Discuss how air occupies space.
- Create a poster highlighting ways to reduce air pollution.



Critical Thinking

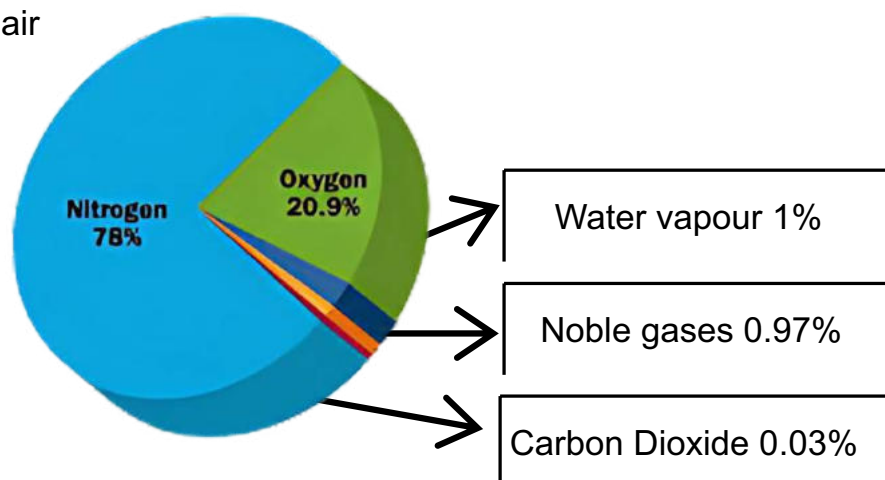
- Why is it important to reduce air pollution, and how can students contribute to this effort?



AIR AND ITS COMPOSITION

Answer the questions.

1. The figure below shows a pie chart representing the composition of the air



a) Is air an element, a compound or mixture?

b) What percentage of air is nitrogen?

c) What percentage of air is oxygen?

d) What percentage of air is carbon dioxide?

e) What other gases are present in air?

f) What are the noble gases found in air?

g) Is the composition shown in the pie chart the same in different parts of the world?